TravelTide rewards pilot

Executive summary | Prepared by Miguel | Analysis reference: 28 July 2023

Business question

TravelTide's brief links underdeveloped customer experience to poor retention. The proposed rewards program should give existing customers a relevant reason to join and return. This analysis asks how observed behavior can guide member value without committing to blanket discounts.

Analysis and key findings

Four activity segments cover all 5,782 selected customers. K-means uses days since the last recorded platform session and noncanceled trip-booking frequency in a defined recent window. The table presents group medians, not individual thresholds or verified customer motives.

Segment	Customers	Activity medians	Primary proposal
Regulars	2,041	17 days; 3 bookings	Partner-hotel late checkout or upgrade, subject to availability
Reconnectors	2,239	52 days; 3 bookings	Capped next-booking voucher with proposed partner cost-sharing
Explorers	666	20 days; 1 booking	Member planning tools: saved options and price alerts
Occasional Visitors	836	61 days; 1 booking	Preference-led welcome with approved benefit choices

Regulars and Reconnectors comprise 4,280 customers: 74% of the study cohort. Both have a median of three recent bookings, but different engagement recency. This supports testing different invitations; it does not establish profitability or response.

Recommendation and decision

Approve feasibility work and a bounded pilot, not a full rollout. Confirm service capability, participating partners, funding, eligibility, cancellation terms and maximum financial exposure. Compare relevant and general invitations through customer-level random assignment, with comparable program access. Measure enrollment first, then subsequent bookings, fulfillment and costs.

Economic and evidence safeguards

Benefits are testable proposals, not verified preferences. The original five perks have overlapping applicability signals but no observed preference or causal-response evidence. A preference-led welcome is proposed where the evidence does not justify a specific expensive reward. Booking expenditure is not TravelTide revenue or profit. Scale only when incremental outcomes and actual economics support it.

Scope: more than seven sessions from 5 January 2023; recent booking window: 5 January–28 July 2023. Earlier available history is retained for profiling. The 74% is not a share of all TravelTide customers.

Detailed report — evidence and method

1. Scope and data preparation

The analysis uses TravelTide's users, sessions, flights and hotels exports. Eligibility selects customers with more than seven sessions from 5 January 2023. Available earlier history is retained rather than discarded. The final cohort contains 5,782 customers, including customers without bookings or usable expenditure. The reference date is 28 July 2023, the latest observed session date in the export.

The pipeline preserves raw inputs and builds separate processed customer tables. Corrections include combining flight and hotel expenditure rather than adding flight cost twice; using a fixed historical date for age and recency; validating calendar-night differences; and linking discount-purchase counts to the product actually booked. Unknown expenditure or duration is not silently converted into a valid zero. Cancellation handling distinguishes never-booked customers from those with only canceled bookings.

2. Final features and observation windows

Engagement recency: days from a customer's last recorded platform session to the fixed reference date. This is not booking recency. Frequency: noncanceled trip bookings recorded from 5 January through 28 July 2023. Noncanceled does not confirm completed travel; a travel date may lie after the observation period. Booking frequency is not the sum of flight and hotel product counts.

A common recent window makes comparisons explicit, but later joiners still have shorter exposure. Earlier-history features and recent-window features are separately labeled. Segment names describe relative activity patterns; they do not establish loyalty, churn, planning difficulty or traveler identity.

3. Exploration and final model choice

RFM and LRFM were explored on the same 4,854 customers with complete eligible inputs. RFM uses booking recency, recent booking frequency and estimated booking expenditure. LRFM adds the observed first-to-last noncanceled-booking span, not customer tenure. These analyses remain benchmarks and profiling tools.

The final activity model was selected for full-cohort coverage and understandable marketing profiles, without inventing missing booking-recency or spending values. Both inputs use log1p transformation and population-standard-deviation scaling. K-means uses k-means++ initialization, ten starts and sensitivity runs with seeds 42, 43 and 44.

Within this two-feature family, two through six groups were tested. Four had the highest silhouette on a fixed 1,000-customer sample: approximately 0.398. Minimum seed-agreement adjusted Rand index was approximately 0.976. These are internal separation and initialization-stability checks, not evidence of future retention or benefit response. They do not prove that this model is universally superior to RFM/LRFM.

Detailed report — benefit evidence

4. Assessment of Elena’s original five perks

Historical behavior can indicate where a perk might be usable, but cannot reveal a “favorite” benefit with certainty. The evidence below uses available full-history features. The flags overlap: they are not five exclusive customer segments and must not be summed.

Original perk hypothesis	Observed signal	What the evidence supports
Free checked baggage	Baggage use: 3,870 customers	Applicability for observed baggage users; no verified preference or airline funding.
Exclusive discounts	Discounted purchases: 2,671	Past discount use; not causal price sensitivity or incremental demand.
Free cancellation	Cancellation observed: 596	Investigate flexibility; actual fees and desired cancellation terms are unknown.
Free hotel meal	Hotel bookings: 5,180	Possible applicability, not observed meal preferences.
Free hotel night with flight	Flight and hotel bookings: 4,718	Potential package applicability; not proof of a free-night preference or viable cost.

5. Why the final proposals differ

An earlier provisional policy mapped customers to Elena’s five perks. It remains exploratory evidence, not the final business proposal. The final design uses activity-based invitations and commercially constrained member value. It does not claim to have empirically discovered optimal perks. This is a recommendation to refine the original hypotheses, not validation of all five.

Regulars: 2,036 of 2,041 customers have observed hotel bookings. A partner stay enhancement is therefore a plausible applicable proposal, not a verified preference. The five customers without observed hotel bookings receive the planning-service alternative. In the same available history, 2,000 Regulars have both flight and hotel bookings; these need not refer to the same trip.

Reconnectors: propose a voucher of capped value for a future qualifying booking. Less recent engagement creates a contact opportunity; it does not diagnose churn or prove price sensitivity. Partner participation and the risk of subsidizing bookings that would occur anyway must be evaluated.

Explorers: propose saved travel options and price alerts before monetary incentives. These functions must add value beyond existing search. Technical capability, development and operating costs, and notification consent require review. Planning friction is a hypothesis, not an observed cause.

Occasional Visitors: propose a preference-led welcome with a selection of approved, budgeted benefits. This assigns a welcome policy, not a verified single favorite perk. Concrete choices and customer response must be established before rollout.

Detailed report — economics and handoff

6. Economic conditions before launch

The shared checks for all four segments are understandable customer value, feasible delivery and bounded cost exposure. Specify who pays, who qualifies, how fulfillment works and maximum obligations. Hotel benefits require participating properties, availability and redemption limits; unused hotel capacity is not automatically free. Voucher proposals require a fixed budget, approved value and eligibility, partner cost-sharing where feasible, and explicit cancellation and reimbursement terms. Capped value does not itself establish expiry or legal compliance.

Planning services require development and operating budgets, not an assumption of zero cost. Preference-led choices must be genuinely available and approved. The dataset supplies no partner contracts, benefit costs, commission rates or margins; a quantified profit or return-on-investment forecast is therefore not justified.

7. Controlled pilot and scale decision

First verify feasibility and funding. Then randomly assign customers within segments to a behavior-relevant invitation or a general invitation, with comparable program access. This tests the invitation strategy. To assess the incremental effect of the overall program, use an appropriate no-offer or delayed-offer comparison. Specify the outcome window and success criteria in advance.

Evaluate all randomly assigned customers, not only email openers or redeemers. Track enrollment, subsequent noncanceled bookings, usage, fulfillment and costs. Enrollment is not retention, and redemption is not incrementality. Assess actual contribution only when TravelTide's revenue or margin inputs and delivery costs are available; gross booking value alone is insufficient. Expand only when the measured effects and economics support the decision.

8. Reproducible artifacts and assignment coverage

The final customer-level output, customer_segments_business_aligned.csv, contains 5,782 rows and 5,782 unique user IDs, with no missing proposed-benefit labels. It records segment names and descriptions, proposed member value, applicability fallback and economic/feasibility conditions. These are proposed assignments, not fulfilled rewards or proven preferences.

Supporting outputs are named_segment_profiles.csv (original-unit segment profiles), segment_benefit_summary.csv (reconciled proposal counts), full-cohort candidate diagnostics and customer-level original-perk evidence. The final proposal is built in 09_update_names_and_business_proposal.ipynb. Earlier notebooks preserve feature construction, exploratory RFM/LRFM, perk evidence and model comparisons.

The submission folder should identify a canonical execution order in README.md, document data access, environment requirements, inputs, outputs and the relationship to classroom examples. Exploratory versions should remain distinguishable from final outputs. Repository cleanup is an organization task, not a change to this report's analytical results.

Decision requested

Authorize feasibility work and a bounded learning pilot—not a full launch. The contribution is a transparent analytical workflow and a cost-conscious, testable rewards strategy. No measured retention improvement, confirmed preference or profitable uplift is claimed.